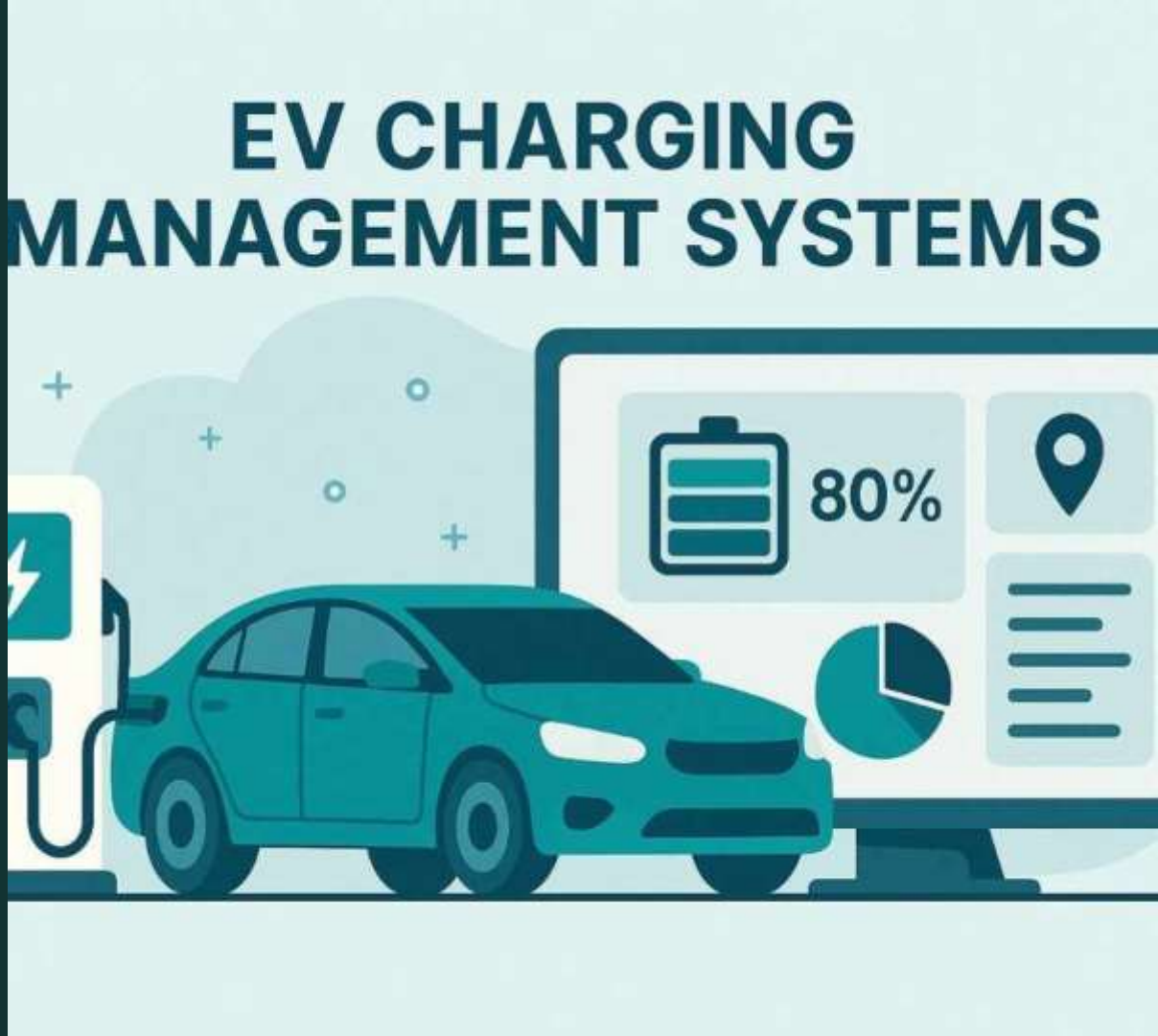
EV Charging Station Management System

5-layer architecture for reliable, scalable, and intelligent charging operations



Core Modules in the Application Layer



Users

Customer and admin account management across the platform.



Vehicles

EV registration, profiles, and real-time tracking.



Stations

Charging station inventory, status, and availability management.



Booking

Reservation and scheduling system for seamless planning.



Charging

Real-time charging session monitoring and control.



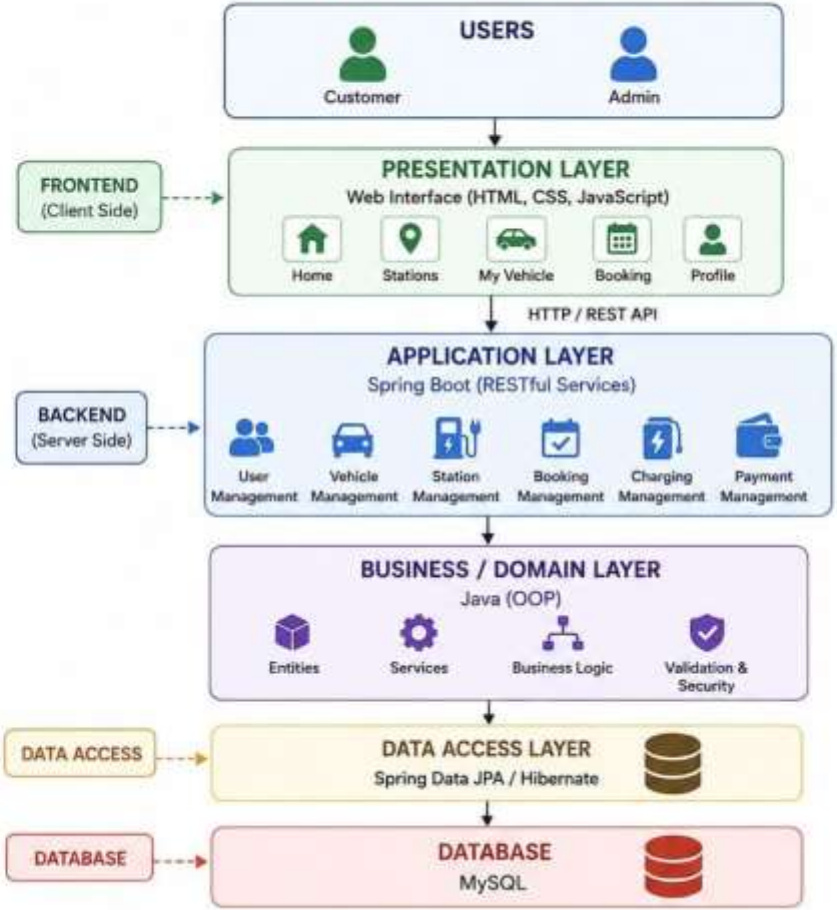
Payments

Transaction processing, billing, and financial records.

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ARCHITECTURAL DIAGRAM

EV Charging Station Management System



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EV Charging Station Management System

A structured walkthrough of the 5-layer architecture powering scalable, secure, and maintainable EV infrastructure management.

Presentation Layer

HTML, CSS, JavaScript

- Home
- Charging Stations
- My Vehicle
- Booking
- Profile

Application Layer

Spring Boot, REST APIs

- User Management
- Vehicle Management
- Station Management
- Booking
- Charging
- Payment Management

Business/Domain Layer

Java OOP logic

- Entities
- Services
- Validation
- Security

Data Access Layer

Spring Data JPA, Hibernate

Database Layer

MySQL

- Users
- Vehicles
- Charging Stations
- Bookings
- Other required system data

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KEY BENEFITS

Why This Architecture Works

The layered design of the EV Charging Station Management System creates a strong foundation for building a platform that is secure, scalable, and easy to evolve.



Separation of Concerns

Each layer has distinct responsibilities, making the system cleaner, easier to understand, and simpler to manage.



Security

Validation and protection at the Business Layer help safeguard data, users, and charging operations.



Maintainability

A modular structure enables easy updates, faster troubleshooting, and lower long-term development effort.



Scalability

The layered approach supports growth as more vehicles, stations, users, and transactions are added.



Reliability

Structured interaction between layers ensures consistent behavior and dependable performance.



Flexibility

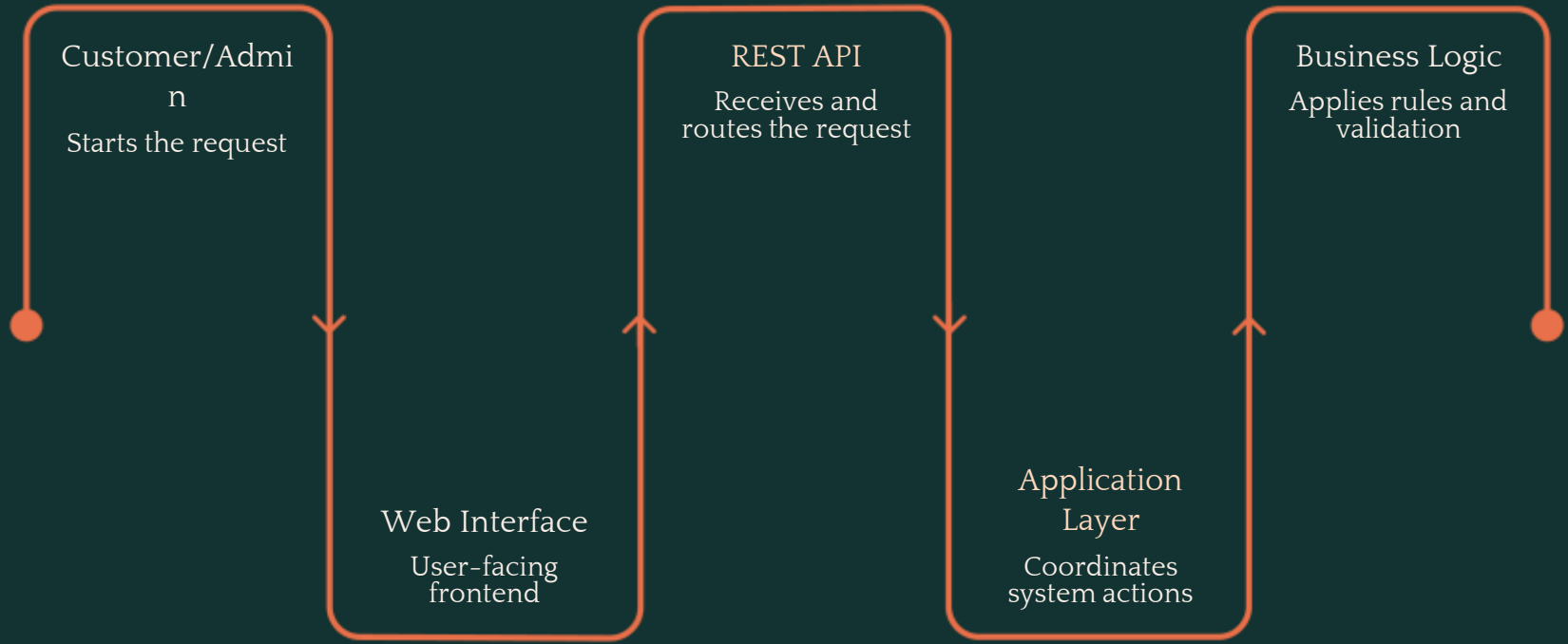
Individual layers can be modified or extended without disrupting the entire system.

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OVERALL WORKFLOW

How the EV Charging Station Management System Works End-to-End

The system follows a clear request-response cycle: a user action starts at the frontend, moves through the application, business, and data layers, reaches MySQL, and then returns back up with the result.



This flow shows how a typical operation is handled: the customer or admin begins in the web interface, the REST API receives the request, the application and business layers process it, JPA/Hibernate manages persistence, and MySQL stores or returns the data before the response is sent back to the user.

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